Job Market Intelligence Platform

Project overview · architecture, decisions and results · Sbusiso Shinga · 20 October 2026

The question: what technologies are employers actually asking for?

Students and training providers often choose what to learn from guesswork. This platform collects real job postings, extracts the skills each one asks for, and turns that into rankings, trends and salaries.

1. Results at a glance

Warehouse snapshot on 20 October 2026, after the batch pipeline (public APIs plus two simulated feeds).

Job postings	Companies	Skills	Job–skill links	Postings with a salary
6,720	2,487	135	22,885	939

Skill	Jobs	Share	Skill	Jobs	Share
Python	1,709	39%	LLMs	613	14%
Machine Learning	981	22%	Kubernetes	612	14%
SQL	920	21%	Azure	575	13%
AWS	873	20%	Docker	487	11%
CI/CD	759	17%	GCP	484	11%

Among 193 Data Engineering postings, the top skills are Python (142, 74%), SQL (135, 70%), Data Quality (86), AWS (85) and dbt (80). Where at least 30 postings state a salary, the highest medians are OpenAI API (US$197,500), Go (US$182,000) and Generative AI (US$177,500).

Two feeds are simulated because no public source offers them: a partner SFTP drop (139 jobs) and a recruitment CRM (92 jobs). Their companies are fictional. Filter with `source not in ('partner_sftp', 'crm_postgres')` for market-only analysis. Arbeitnow contributes the largest share of real postings and publishes no salaries, which is why only 939 of 6,720 rows have a salary.

2. Architecture

One Kafka topic feeds a batch path (accurate, historical) and a speed path (live within seconds). A third path streams changes from the operational CRM.

Stage	What it does
Sources	Six public job APIs. A partner drops `jobs_YYYY-MM-DD.csv` on SFTP. The recruitment desk's CRM is a separate PostgreSQL database.
Ingestion	Python maps each record onto one contract. Invalid rows go to `jobs.dlq`. SFTP files move to archive only after Kafka confirms every row.
Kafka	Topic `jobs.raw`. Offsets are committed after the lake file is written.
Lake	RustFS (S3). Raw zone is gzipped JSONL, partitioned by source and date. Spark writes a Parquet curated zone.

Spark	Dedupes on `job_id`, parses salary and experience, extracts skills, and fails the run if the quality gate finds bad rows.
Warehouse	PostgreSQL. Idempotent upsert into `core`. dbt builds staging, an intermediate model, and marts, with tests.
Speed layer	A consumer updates a Redis skill leaderboard as events arrive. FastAPI serves it at `/skills/realtime`.
CDC	Debezium tails `crm.vacancies` (logical replication) into Kafka. A consumer keeps the newest changes in Redis. `GET /jobs/latest` serves them.
Orchestration	Airflow runs partner drop and CRM activity, then ingest, lake, Spark and dbt, twice a day. Each task is a container.

3. Decisions worth defending

- **Raw first.** The API payload is stored untouched, so any transform can be replayed by re-running Spark.
- **At least once, made safe.** Spark dedupes. The upsert rewrites a row only when values changed, so a re-run is free.
- **The CRM is not the warehouse.** It is a normalised OLTP database with constraints and an `updated_at` trigger. The platform reads it in a read-only session over an overlapping lookback window.
- **CDC is on the source, not on the mart.** Recruiters' changes show up in seconds. The batch extract remains the complete, replayable copy.
- **One image.** Ingestion, consumers, Spark and the API share one image. Airflow only orchestrates.

4. How a ten-minute demo runs

1. Show the dashboard: Python is in 39% of postings.
2. `make partner-drop` and `make crm-activity`, then the topics in Kafka UI.
3. `make pipeline`, then `make top-skills`. Run it again: unchanged rows are not rewritten.
4. `make cdc`, change the CRM again, then `curl localhost:8000/jobs/latest`.
5. Open the Airflow DAG `job_market_pipeline`.

Repository: https://github.com/Sbusiso-tech/job-market-intelligence-platform